



JADARD

JD9853

Internal Command

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1. Revision History

Version	Date	Description of modification
0.00	2023/07/14	New setup
0.01	2023/08/18	Correcting description for P0, DFh
0.02	2023/09/20	Correcting description to P0, C4h
0.03	2023/11/28	Optimizing OTP programming flow
0.04	2024/1/26	Optimizing OTP programming flow

2. JADARD COMMAND LIST

2.1. Jadard page 0 command

Index	Operation	R/W	D7	D6	D5	D4	D3	D2	D1	D0	Default	
	Code										(Hex)	
B0	SEQUENCE_CTRL	R/W	-	-	-	-	DC_UP_MODE[1:0]		DC_DN_MODE[1:0]		00	
		R/W	-	-	-	-	TOP[1:0]		T8P[1:0]		0A	
		R	-	-	-	-	-	-	-	POW_ON_STATE	00	
B2	VGPOW_SE T	R/W	-	VGPOW_OFFSET[6:0]							28	
		R	-	-	-	-	-	-	VGPOW_SET_OTP_TIM ES[1:0]		00	
B7	GAMMA_SE T	R/W	-	VGSP_S[6:0]							09	
		R/W	-	VGMP_S[6:0]							29	
		R/W	-	VGSN_S[6:0]							09	
		R/W	-	VGMN_S[6:0]							29	
		R	-	-	-	-	-	-	GAMMA_SET_OTP_TIM ES[1:0]		00	
B8	OTP_SET	R/W	-	-	VPP_DT[1:0]		-	-	VPP_VGHZ_RT[1:0]		33	
		R/W	VPP_VGHZ_NC	VPP_VGHZ_S[2:0]			VPP_PWRC LK_DIV2	VPP_VGHZ_CLK[2:0]			53	
		R/W	-	-	VPP_AVDDZ_S[1:0]		-	VPP_MVZ_S_CLK[2:0]			07	
B9	POW_CTRL	R/W	-	AP[2:0]			-	MVZ_EN	-	VGM_EN	35	
		R/W	AVDDZ_NC	AVEEZ_NC	VGHZ_NC	VGLZ_NC	-	-	VGHZ_EN	VGLZ_EN	03	
BB	DCDC_SEL	R/W	-	VGHZ_S[2:0]			PWRCLK_D IV2	VGLZ_S[2:0]			3C	
		R/W	VGHZ_RT[1:0]		AVDDZ_S[1:0]		VGLZ_RT[1:0]		AVEEZ_S[1:0]		95	
		R/W	-	VGHZ_CLK[2:0]			-	VGLZ_CLK[2:0]			77	
		R/W	-	MVZ_S_CLK[2:0]			-	MVZ_G_CLK[2:0]			77	
		R/W	-	MVZ_NS_CLK[2:0]			HV_CLKP[1:0]		MV_CLKP[1:0]		7A	
		R/W	VGHZ_CLKE[1:0]		VGLZ_CLKE[1:0]		MVZ_S_CLKE[1:0]		MVZ_NS_CLKE[1:0]		FF	
BC	VDDD_CTRL	R/W	-	AVSSN_EN	AVSSN_S[1:0]		-	VDDD_S[2:0]			73	
BD	SETRGBIF	R/W	-	-	BLK_OPT	DEM	VSPL	HSPL	DEPL	PCPL	22	
		R/W	-	-	RGB_SEL[1:0]		-	-	-	-	00	
BE	GAS_CTRL	R/W	-	-	GAS_SLPI N_EN	-	GAS_VCI EN	GAS_IOVC C_EN	GAS_BLK_NUM[1:0]		2E	
		R/W	GAS_DBS_SEN	GAS_DBS_GEN	-	GAS_DBS_LTH[4:0]						8C
		R/W	-	-	DC_GAS_VGH_EN	DC_GAS_VGL_EN	-	-	DC_GAS_MVZ_EN	-	20	
C0	SETSTBA	R/W	-	GAP[2:0]			SAP[3:0]				22	
		R/W	SAP2_GAP 2_EN	GAP2[2:0]			SAP2[3:0]				22	
C1	SETPANEL	R/W	-	-	-	SS_PANEL	GS_PANEL	REV_PANE L	CFHR	-	00	
C2	SET_BIST	R/W	-	LNPERLVL[1:0]		-	TEST_PAT EN	TEST_PATTERN[2:0]			00	

C3	SETRGBCY C	R/W	-	RGB_INV_P[1:0]		RGB_INV_I[1:0]		IDLE_TYPE	RGB_INV_NP[1:0]	7F	
		R/W	-	-	-	-	RGB_GND[3:0]			07	
		R/W	-	-	-	RGB_EQ1[4:0]				0E	
		R/W	-	-	-	RGB_EQ2[4:0]				0E	
		R/W	RGB_CHGEN_OFF[1:0]		RGB_CHGEN_ON[5:0]					C8	
		R/W	RGB_OFF[7:0]							71	
		R/W	GAOPOFF_1ST_OFF	GAOPOFF_POL	GAOPOFFEN	GAOPOFF[4:0]				6C	
		R/W	GAP2_ON[3:0]				SAP2_ON[3:0]			77	
C4	SET_TCON	R/W	-	-	-	-	VBFP_RATIO[1:0]		TE_OPT[1:0]	00	
		R/W	-	TE_DELAY[6:0]							00
		R/W	LN[7:0]							A0	
		R/W	SLT_NP[7:0]							79	
		R/W	-	VFP_NP[6:0]						0E	
		R/W	-	VBP_NP[6:0]						0A	
		R/W	-	HBP_NP[6:0]						16	
		R/W	SLT_I[7:0]							79	
		R/W	-	VFP_I[6:0]						0E	
		R/W	-	VBP_I[6:0]						0A	
		R/W	-	HBP_I[6:0]						16	
		R/W	HBP_NCK[3:0]				HFP_NCK[3:0]			82	

C8	SET_R_GA MMA	R/W	-	-	RPA15[5:0]	3F
		R/W	-	-	RPA14[5:0]	3E
		R/W	-	-	RPA13[5:0]	23
		R/W	-	-	RPA12[5:0]	11
		R/W	-	-	RPA11[5:0]	13
		R/W	-	-	RPA10[5:0]	14
		R/W	-	-	RPA9[5:0]	13
		R/W	-	-	RPA8[5:0]	17
		R/W	-	-	RPA7[5:0]	1B
		R/W	-	-	RPA6[5:0]	1E
		R/W	-	-	RPA5[5:0]	22
		R/W	-	-	RPA4[5:0]	23
		R/W	-	-	RPA3[5:0]	25
		R/W	-	-	RPA2[5:0]	28
		R/W	-	-	RPA1[5:0]	1F
		R/W	-	-	RPA0[5:0]	0E
		R/W	-	-	RNA15[5:0]	3F
		R/W	-	-	RNA14[5:0]	3E
		R/W	-	-	RNA13[5:0]	23
		R/W	-	-	RNA12[5:0]	10
		R/W	-	-	RNA11[5:0]	12
		R/W	-	-	RNA10[5:0]	14
		R/W	-	-	RNA9[5:0]	11
		R/W	-	-	RNA8[5:0]	12
		R/W	-	-	RNA7[5:0]	14
		R/W	-	-	RNA6[5:0]	17
		R/W	-	-	RNA5[5:0]	18
		R/W	-	-	RNA4[5:0]	16
		R/W	-	-	RNA3[5:0]	18
		R/W	-	-	RNA2[5:0]	18
		R/W	-	-	RNA1[5:0]	18
		R/W	-	-	RNA0[5:0]	0E

D0	SET_GD	RW	-	-	-	-	-	GD_GAS_GATE_CLK	-	GD_LRSW	04	
		RW	-	-	GD_ON[5:0]							06
		RW	GD_OFF[7:0]									6B
		RW	-	-	-	GD_EQ_PTR0[4:0]						07
		RW	-	-	-	GD_EQ_PTR1[4:0]						03
D3	ABE_CTRL	RW	-	ABE_EN	ABE_DIM_EN	LE_ABE_BASE[4:0]					28	
		RW	ABE_DIM_FPS[3:0]				-	-	-	-	16	
		RW	ABE_GATING_EN[7:0]									FF
D7	RAMCTRL	RW	-	CR_OPTION	SPI_2LAN_EN	RP	RM	MLBIT_INV	DM[1:0]		00	
		RW	SCL_PL	EXT_VS_PL	EPF[1:0]		ENDIAN[1:0]		MDT[1:0]		30	
D8	AUTO_DISP_SETTING	RW	-	-	-	-	GON	DTE	DISP_REG[1:0]		08	
		RW	-	-	DISP_SW_OPT[1:0]		-	BLK_DISABLE	BLK_MODE[1:0]		01	
D9	OTP_PROG	RW	-	OTP_INDEX[6:0]							00	
		RW	-	-	-	-	-	-	OTP_INT_VPP	OTP_AUTO_PROG	00	
		R	OTP_RDATA[7:0]									-
		RW	OTP_WDATA[7:0]									00
		RW	-	-	-	-	-	OTP_RLD_PTM[1:0]		-	00	
DD	SET_WD	RW	-	WD_OFF	WD_MODE[1:0]		FBLK_EN	WD_CLK_SEL[2:0]			2C	
		RW	WD_TIMER_TCON[7:0]									A3
		RW	-	-	-	-	-	ESD_TCON_OPT	ESD_SD_OPT	ESD_PRT_EN	00	
		R	-	-	-	-	-	-	WD_FMS	WD_FBLK	00	
DE	SET_PAGE	RW	-	-	-	-	-	-	PAGE[1:0]		00	
DF	SET_PASSWD	RW	PASSWD1[7:0]									00
		RW	PASSWD2[7:0]									00
E8	UP_START_CTRL	RW	UP_KEY1[7:0]									00
		RW	UP_START_EN[7:0]									00
		RW	PWR_KEY[7:0]									00
E9	SETID	RW	-	-	-	-	-	-	-	-	00	
		RW	ID1[7:0]									98
		RW	ID2[7:0]									53
		RW	ID3[7:0]									00
		RW	ID4[7:0]									00
		R	-	-	-	-	-	-	SETID_OTP_TIMES[1:0]		00	

2.2. Jadard page 1 command

Index	Operation	R/W	D7	D6	D5	D4	D3	D2	D1	D0	Default
	Code										(Hex)
BB	SET_GPO	R/W	-	TE_REV	-	-	TE_SEL[3:0]				01
		R/W	-	GPO0_REV	GPO0_OE	-	GPO0_SEL[3:0]				22
		R/W	-	GPO1_REV	GPO1_OE	-	GPO1_SEL[3:0]				23
		R/W	INTERNAL_SEL2[3:0]				INTERNAL_SEL1[3:0]				00
BE	MA_POWER_	R/W	-	-	-	-	VGMP_TES T	VGSP_TES T	VGSN_TES T	VGMM_TES T	0F
C1	SETRGBCY C2	R/W	-	SDPRDUM	SDDUM[1:0]		SDSW[1:0]		SDPORCH[1:0]		10
C4	SET_GAMM A_OP	R/W	-	GAOPENN	GAOPENC	GAOPENP	R1P[1:0]		R0P[1:0]		70
		R/W	-	-	-	GAOP_CL AMP	R1N[1:0]		R0N[1:0]		10
C5	SET_OSCM	R/W	-	-	OSC_DIV[1:0]		-	-	OSCM_ADJ[1:0]		01
		R/W	-	-	-	-	-	-	OSCM_EN	OSCM_BIA S_EN	00
		R/W	OSC_KEY[7:0]								
D7	SET_INH_CH KSUM	R/W	-	INH_CHKSUM_PAGE_S EL[1:0]		SLPIN_RES ET_EN	INH_CHKS UM_OFF	INH_CHKSUM_SEL[2:0]			1A

2.3. Jadard page 0 command Description

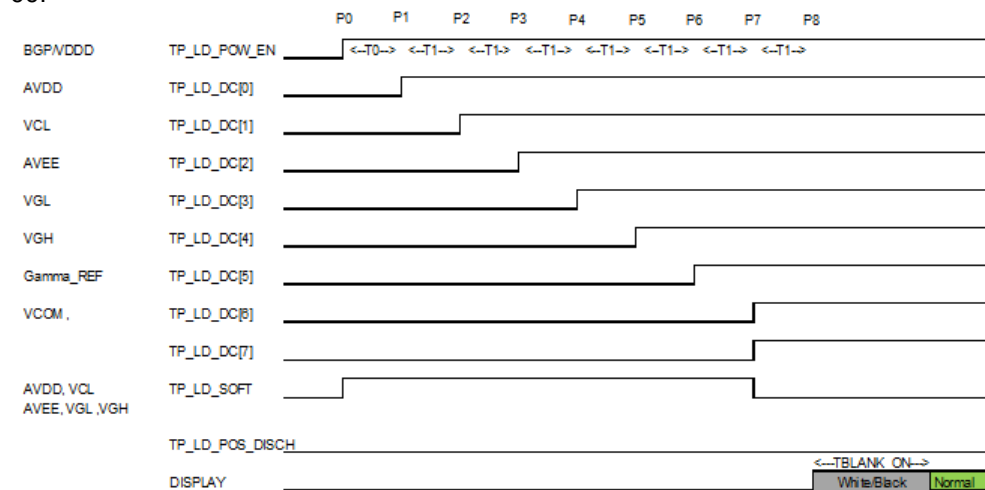
2.3.1. SEQUENCE CTRL: Power on sequence control (Page0 - B0h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	0	1	1	0	0	0	0	B0
Parameter 1	R/W	-	-	-	-	DC_UP_MODE[1:0]		DC_DN_MODE[1:0]		-
Parameter 2	R/W	-	-	-	-	T0P[1:0]		T8P[1:0]		-
Parameter 3	R	-	-	-	-	-	-	-	POW_ON_STATE	-

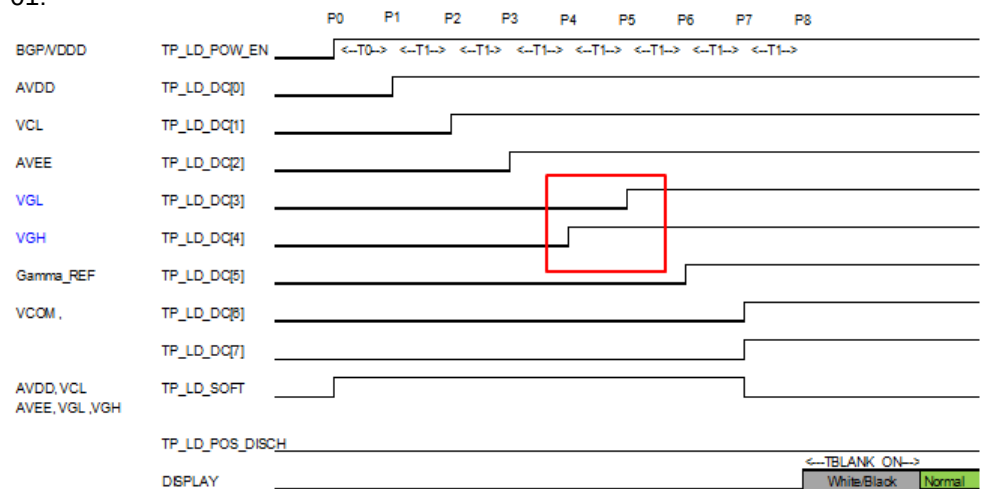
This command is used to set power on-off sequence related setting.

DC_UP_MODE[1:0]: Select the power on sequence timing mode.

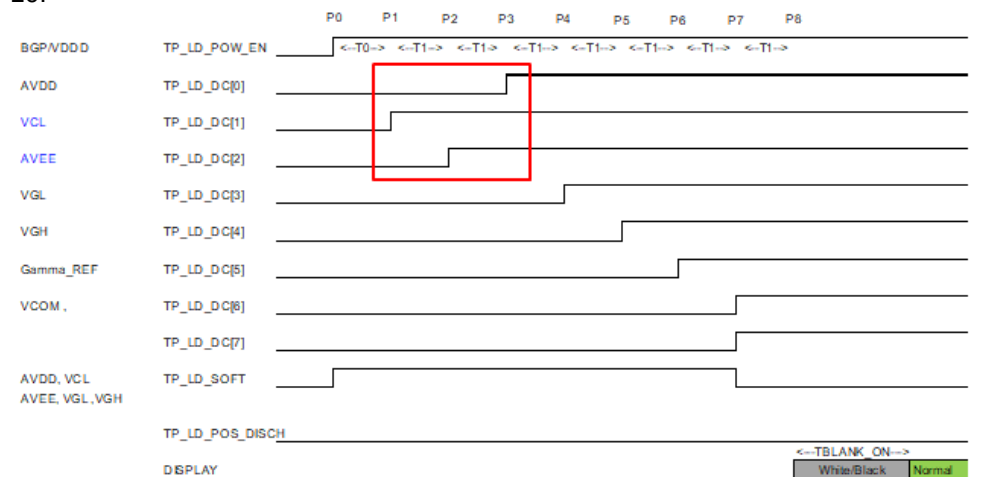
00:



01:

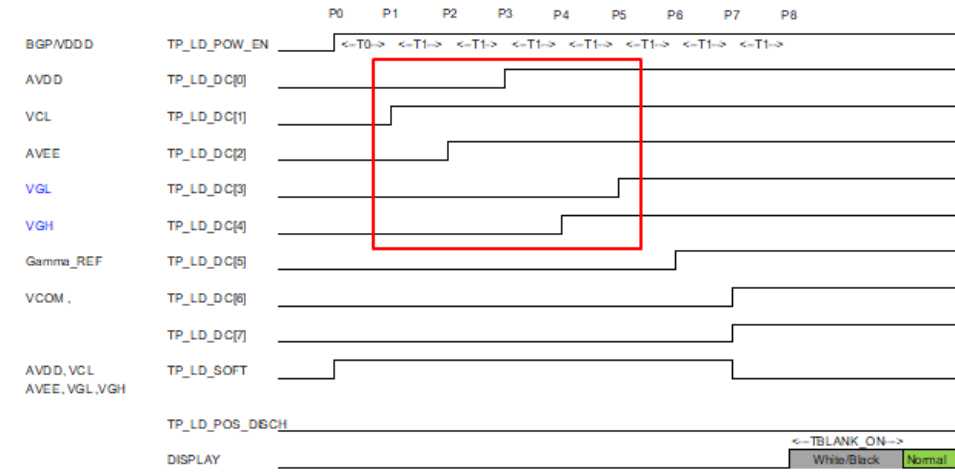


10:



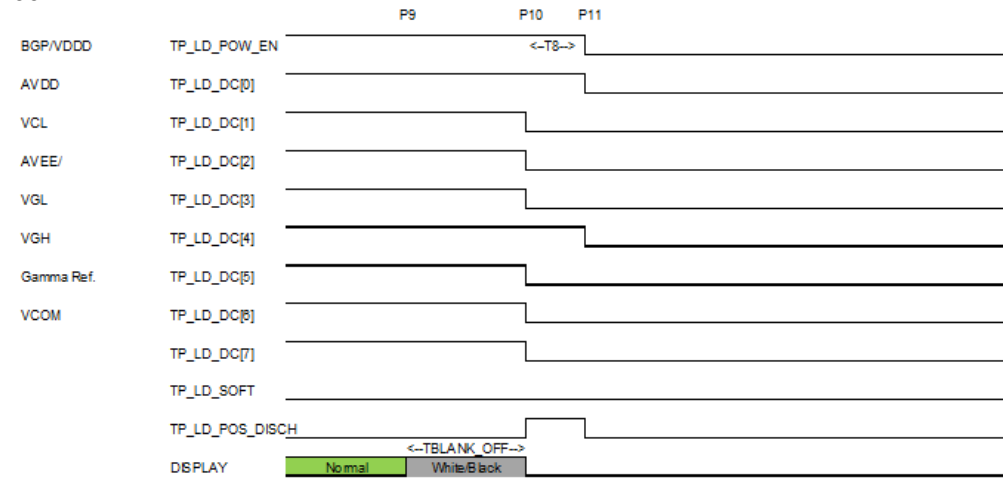
Description

11:

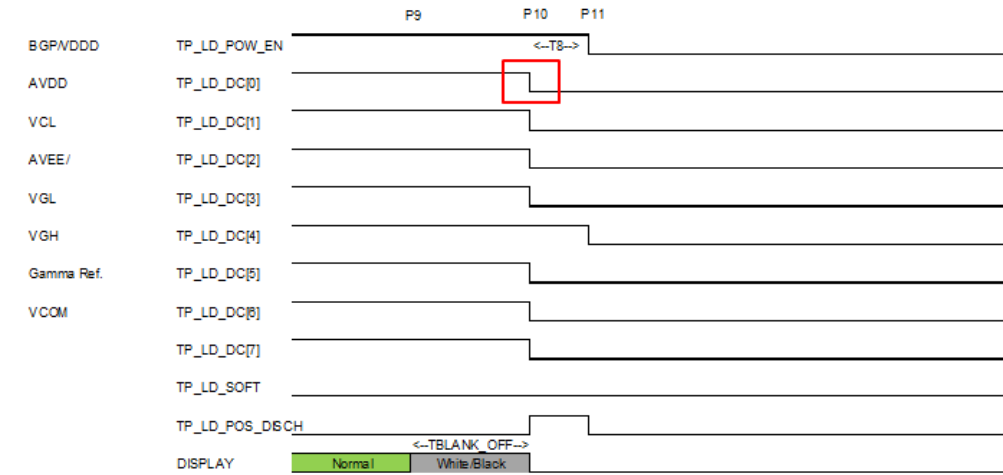


DC_DN_MODE[1:0]: Select the power off sequence timing mode.

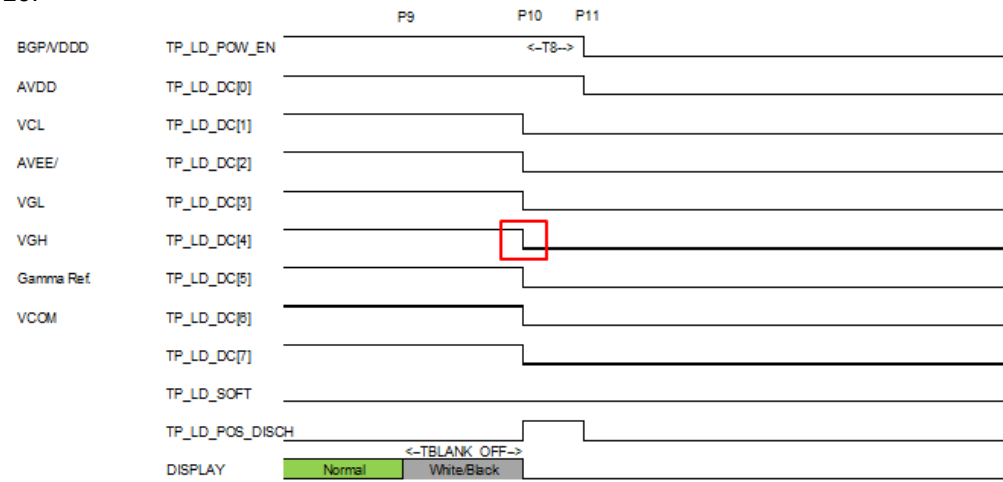
00:



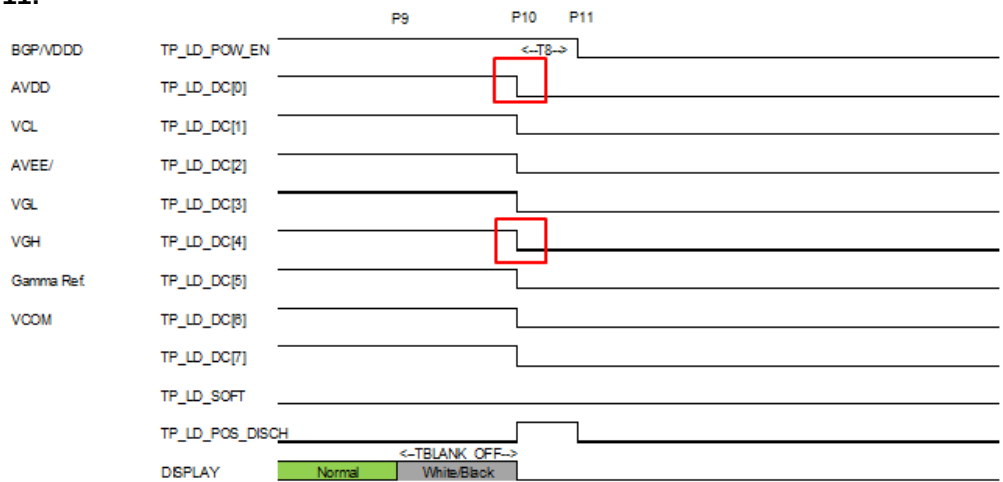
01:



10:



11:



T0P[1:0]: Select the period time for T0.

T8P[1:0]: Select the period time for T8.

Power-on/off Period option		
TxP[1:0]	Period	
0	0.1ms	
1	1.6ms	
2	3.2ms	T0/T8 default
3	6.4ms	

POW_ON_STATE: All power ready when this bit is high.

Restriction

2.3.2. VCOM_SET: Set VCOM voltage for normal scan direction(Page0 - B2h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	0	1	1	0	0	1	0	B2
Parameter 1	R/W		VGPOW_OFFSET[6:0]							-
Parameter 2	R	-	-	-	-	-	-	VGPOW_OTP_TIMES[1:0]		-
Description	This command is used to set VCOM voltage (used to Gamma shift voltage) and read OTP writing times for VCOM. (Gamma setting please refer to page2 BEh)									
	VGPOW_OFFSET[6:0]: Set VCOM voltage when normal scan.									
	VGPOW_OFFSET[6:0]					VCOM(v)				
	00h ~ 0Bh					Inhibit				
	0Ch					-0.300				
	~					-25mv/step				
	28h					-1.000				
	~					-25mv/step				
	64h					-2.500				
	65h ~ 7Fh					Inhibit				
	VGPOW_OTP_TIMES[1:0]: This register just can be written by up.									
	Due to VCOM is belonging to MTP, VGPOW_OTP_TIMES[1:0] is used to record OTP being programmed times of VCOM.									
	VCOM_OTP_TIMES[1:0]					OTP programmed times				
	00					No programmed				
	01					OTP programmed 1 times				
	10					OTP programmed 2 times				
Others					Inhibit					
Restriction	-									

2.3.3. GAMMA_SET: Set positive / negative voltage of Gamma power (Page0 - RB7h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	0	1	1	0	1	1	1	B7
Parameter 1	R/W	-	VGSP_S[6:0]							-
Parameter 2	R/W		VGMP_S[6:0]							-
Parameter 3	R/W	-	VGSN_S[6:0]							-
Parameter 4	R/W		VGMN_S[6:0]							-
Description	This command is used to set postive / Negative reference volatge of Gamma. (Gamma setting please refer to page2 BEh)									
	When Page1,BEh(GAMMA_POWER_TEST) set to 0x0F									
	VGMP_S[7:0]: Set power source volatge of postive polarity.									
	VGMP_S[7:0]		VGMP(V)							
	00h		0							
	01h		3.500							
	~		0.025/step							
	29h		4.500							
	~		0.025/step							
	7Dh		6.600							
7Eh		AVDD								
7Fh		Hiz								
VGMN_S[7:0]: Set power source volatge of negative polarity.										
VGMN_S[7:0]		VGMN(V)								
00h		0								
01h		-2.500								
~		0.025/step								
29h		-3.500								
~		0.025/step								
7Dh		-5.600								
7Eh		AVEE								
7Fh		Hiz								
VGSP_S[6:0]: Set reference volatge of postive polarity.										
VGSP_S[6:0]		VGSP(V)								
00h		0								
01h		0.300								
~		0.025/step								
09h		0.500								
~		0.025/step								
5Fh		2.650								
60~7Eh		2.650								
7Fh		Hiz								
VGSN_S[6:0]: Set reference volatge of negative polarity.										
VGSN_S[7:0]		VGMP(V)								
00h		0								
01h		0.300								
~		0.025/step								
09h		0.500								
~		0.025/step								
5Fh		2.650								
60~7Eh		2.650								
7Fh		Hiz								

VPP_PWRCLK_DIV2: Specify the OSCM frequency

0:OSCM=20MHz

1:OSCM=10MHz

VPP_VGHZ_S[2:0]: When OTP_INT_VPP=1, set VGH voltage level of internal OTP program.

VPP_VGHZ_S[2:0]	VPP_VGHZ Voltage
0	7.4
1	7.6
2	7.8
3	8
4	8.2
5	8.4
6	8.6
7	8.8

VPP_VGHZ_CLK[3:0] :

When OTP_INT_VPP=1, TP_LT_CPCLKZ [5] = VPP_VGHZ_CLK[3:0])

VPP_VGHZ_CLK[2:0] :	Frequency
0	OSCM / 24
1	OSCM / 16
2	OSCM / 12
3	OSCM / 8
4	OSCM / 6
5	OSCM / 4
6	OSCM / 3
7	OSCM / 2

AVDDZ_S[1:0]: Set clamping voltage level of AVDDZ.

AVDDZ_S[1:0]	AVDDZ (V)
0h	6.4
1h	6.6
2h	6.8
3h	inhibit

VPP_MVZ_S_CLK[2:0]: Set frequency of pumping clock of VPP_MVZ_NS.

VPP_VGHZ_CLK[2:0] :	Frequency
0	OSCM / 24
1	OSCM / 16
2	OSCM / 12
3	OSCM / 8
4	OSCM / 6
5	OSCM / 4
6	OSCM / 3
7	OSCM / 2

Restriction

-

2.3.5. POWER_CTRL2: Set power function related setting (Page0 - B9h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	0	1	1	1	0	0	1	B9
Parameter 1	R/W	-	AP[2:0]			-	MVZ_EN	-	VGM_EN	-
Parameter 2	R/W	AVDDZ_NC	AVEEZ_NC	VGHZ_NC	VGLZ_NC	-	-	VGHZ_EN	VGLZ_EN	-
Description	This command is used to control engineer mode of power related setting.									
	AP[2:0] : Adjust the ratio of fixed current for the operatinal amplifier in the power supply circuit.									
	AP[2:0]		Bias Current							
	0h		25%							
	1h		50%							
	2h		75%							
	3h		100%							
	4h		125%							
	5h		150%							
	6h		175%							
	7h		200%							
	MVZ_EN : Control Source/Gamma MV charge pump 0: disable 1: enable									
	VGM_EN : Control Gamma power regulator output 0: disable 1: enable									
	AVDDZ_NC : Set AVDDZ output state 0: clamping 1: no Clamping									
	AVEEZ_NC : Set AVEEZ output state 0: clamping 1: no Clamping									
	VGHZ_NC : Set VGHZ output state. 0: clamping 1: no Clamping									
	VGLZ_NC : Set VGLZ output state 0: clamping 1: no Clamping									
VGHZ_EN : Control VGHZ DC/DC circuit On/Off. 0: disable 1: enable										
VGLZ_EN : Control VGLZ DC/DC circuit On/Off. 0: disable 1: enable										
Restriction										

2.3.6. DCDC_SEL: Power mode and charge pump related setting (Page0 - BBh)

CMD	ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command		R/W	1	0	1	1	1	0	1	1	BB
Parameter 1		R/W		VGHZ_S[2:0]			PWR CLK_ DIV2	VGLZ_S[2:0]			-
Parameter 2		R/W	VGHZ_RT[1:0]		AVDDZ_S[1:0]		VGLZ_RT[1:0]		AVEEZ_S[1:0]		
Parameter 3		R/W	-	VGHZ_CLK[2:0]			-	VGLZ_CLK[2:0]			
Parameter 4		R/W	-	MVZ_S_CLK[2:0]			-	MVZ_G_CLK[2:0]			
Parameter 5		R/W	-	MVZ_NS_CLK[2:0]			HV_CLKP[1:0]		MV_CLKP[1:0]		
Parameter 6		R/W	VGHZ_CLKE[1:0]		VGLZ_CLKE[1:0]		MVZ_S_CLKE[1:0]		MVZ_NS_CLKE[1:0]		
Description	This command is used to set charge pump related setting. Which include pumping frequency and ratio of voltage .										
	VGHZ_S[2:0]: Set clamping voltage level of VGLZ										
	VGHZ_S[2:0]		VGHZ Voltage								
	0		11								
	1		12								
	2		12.6								
	3		13.2								
	4		13.8								
	5		14.4								
	6		15								
7		16									
VGLZ_S[2:0]: Set clamping voltage level of VGLZ											
VGHZ_S[2:0]		VGHZ Voltage									
0		-7									
1		-7.75									
2		-8.5									
3		-9.25									
4		-10									
5		-10.75									
6		-11.5									
7		-12.25									
PWRCLK_DIV2: Specify the OSCM frequency 0:OSCM=20MHz 1:OSCM=10MHz											
VGHZ_RT: Specify the charge pump ratio of VGHZ.											
VGH_Z_RT[1:0]		VGHZ charge pump									
0		3*VCIPX2									
1		AVDD+2*VCIPX2									
2		2*AVDD+VCIPX2									
3		3*AVDD									
VGLZ_RT: Specify the charge pump ratio of VGLZ.											
VGL_Z_RT[1:0]		VGLZ charge pump									
0		2*VCL – 2*VCI									
1		3*VCL – 2*VCI									
2		AVEE+VCL – 2*VCI									
3		AVEE+2*VCL – 2*VCI									

AVDDZ_S[1:0]: Set clamping voltage level of AVDDZ.

AVDDZ_S[1:0]	AVDDZ (V)
0h	6.4
1h	6.6
2h	6.8
3h	inhibit

AVEEZ_S[1:0]: Set clamping voltage level of AVEEZ_S.

AVEEZ_S[1:0]	AVEEZ (V)
0h	-4.2
1h	-4.4
2h	-4.6
3h	-4.8

VGHZ_CLK[2:0]: Set frequency of pumping clock of VGHZ under normal mode.

VGLZ_CLK[2:0]: Set frequency of pumping clock of VGLZ under normal mode.

MVZ_S_CLK[2:0]: Set frequency of pumping clock of MVZ_S under normal mode.

MVZ_NS_CLK[2:0]: Set frequency of pumping clock of MVZ_NS under normal mode.

MVZ_G_CLK[2:0]: Set frequency of pumping clock of MVZ_G under normal mode.

MVZ_S_CLKE[2:0]: Set frequency of pumping clock of MVZ_S under EQ mode.

MVZ_NS_CLKE[2:0]: Set frequency of pumping clock of MVZ_NS under EQ mode.

VGHZ_CLKE[2:0]: Set frequency of pumping clock of VGHZ under EQ mode.

VGLZ_CLKE[2:0]: Set frequency of pumping clock of VGLZ under EQ mode.

Register		Register	APR Pin name
MVZ_S_CLK[2:0]		MVZ_S_CLKE[1:0]	TP_LT_CPCLKZ_S
MVZ_NS_CLK[2:0]		MVZ_NS_CLKE[1:0]	TP_LT_CPCLKZ_NS
0h	OSCM_D24	0h~3h	OSCM_D24
1h	OSCM_D16	0h~3h	OSCM_D16
2h	OSCM_D12	0h~3h	OSCM_D12
3h	OSCM_D8	0h~3h	OSCM_D8
4h	OSCM_D6	0h	OSCM_D16
		1h	OSCM_D12
		2h	OSCM_D8
		3h	OSCM_D6
5h	OSCM_D4	0h	OSCM_D12
		1h	OSCM_D8
		2h	OSCM_D6
		3h	OSCM_D4
6h	OSCM_D3	0h	OSCM_D8
		1h	OSCM_D6
		2h	OSCM_D4
		3h	OSCM_D3
7h	OSCM_D2	0h	OSCM_D6
		1h	OSCM_D4
		2h	OSCM_D3
		3h	OSCM_D2

Register	
MVZ_G_CLK[2:0]	
0h	OSCM_D24
1h	OSCM_D16
2h	OSCM_D12
3h	OSCM_D8
4h	OSCM_D6
5h	OSCM_D4
6h	OSCM_D3
7h	OSCM_D2

Register		Register	APR Pin name
VGHZ_CLK[2:0]		VGHZ_CLKE[2:0]	TP_LT_CPCLKZ3
VGLZ_CLK[2:0]		VGLZ_CLKE[2:0]	TP_LT_CPCLKZ4
0h	OSCM_D96	0h~3h	OSCM_D96
1h	OSCM_D64	0h~3h	OSCM_D64
2h	OSCM_D48	0h~3h	OSCM_D48
3h	OSCM_D32	0h~3h	OSCM_D32
4h	OSCM_D24	0h	OSCM_D64
		1h	OSCM_D48
		2h	OSCM_D32
		3h	OSCM_D24
5h	OSCM_D16	0h	OSCM_D48
		1h	OSCM_D32
		2h	OSCM_D24
		3h	OSCM_D16
6h	OSCM_D12	0h	OSCM_D32
		1h	OSCM_D24
		2h	OSCM_D16
		3h	OSCM_D12
7h	OSCM_D8	0h	OSCM_D24
		1h	OSCM_D16
		2h	OSCM_D12
		3h	OSCM_D8

HV_CLKP[1:0]: Set frequency of pumping clock of HV under soft start.

MV_CLKP[1:0]: Set frequency of pumping clock of MV under soft start.

HV_CLKP[1:0]	TP_LT_CPCLKZ3 TP_LT_CPCLKZ4
0h	MV_CLK[2:0] period X1
1h	MV_CLK[2:0] period X2
2h	MV_CLK[2:0] period X4
3h	MV_CLK[2:0] period X8

MV_CLKP[1:0]	TP_LT_MVZ_S TP_LT_MVZ_NS TP_LT_MVZ_G
0h	MV_CLK[2:0] period X1
1h	MV_CLK[2:0] period X2
2h	MV_CLK[2:0] period X4
3h	MV_CLK[2:0] period X8

Restriction

2.3.7. VDDD_CTRL: Control logic voltage setting (Page0 - BCh)

CMD/Pas	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	0	1	1	1	1	0	0	BC
Parameter 1	R/W	-	AVSS N EN	AVSSN_S[1:0]		-	VDDD_S[2:0]			-
Description	This command is used to control internal logic voltage setting.									
	VDDD_S[2:0]: Set VDDD voltage.									
	VDDD_S[2:0]		VDDD (V)							
	0h		1.434							
	1h		1.456							
	2h		1.478							
	3h		1.501							
	4h		1.523							
	5h		1.546							
	6h		1.568							
	7h		1.590							
	AVSSN_EN:Enable/disable AVSSN regulator.									
	AVSSN_S[1:0]:Set AVSSN voltage setting.									
	AVSSN_S[1:0]		AVSSN (V)							
	0h		2.0							
1h		1.9								
2h		1.8								
3h		1.7								
Restriction	-									

2.3.8. SETRGBIF: (Page0 - BDh)

[illegible]

2.3.9. GAS CTRL: GAS function control (Page0 - BEh)

[illegible]

2.3.10.SETSTBA: Set Source Output driving ability (Page0 - C0h)

CMD/Pas	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	0	0	0	0	0	C0
Parameter 1	R/W	-	GAP[2:0]			SAP[3:0]				-
Parameter 2	R/W	GAP2 SAP2 _EN	GAP2[2:0]			SAP2[3:0]				
Description	This command is used to set internal OTP program related setting.									
	GAP2SAP2_EN: GAP2, SAP2 enable									
	.									
	GAP[2:0]: Set Gamma OP output driving ability.									
	GAP2[2:0]: Set Gamma OP output driving ability during idle mode									
	GAP[2:0]/ GAP2[2:0]		Bias current (uA)							
	000		1.79							
	001		3.12							
	010		4.31							
	011		5.43							
100		6.5								
101		7.54								
110		8.55								
111		9.54								
SAP[3:0]: Set channel OP output driving ability of general pattern.										
SAP2[3:0]: Set channel OP output driving ability for power saving										
SAP[3:0]		Bias current (uA)		SAP[3:0]		Bias current (uA)				
0000		0.4		1000		0.3				
0001		0.75		1001		0.58				
0010		1.1		1010		0.85				
0011		1.44		1011		1.114				
0100		1.78		1100		1.37				
0101		2.11		1101		1.63				
0110		2.45		1110		1.88				
0111		2.75		1111		2.13				
Restriction	-									

2.3.11.SETPANEL: Set Panel relate register (Page0 - C1h)

CMD/Pas	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	0	0	0	0	1	C1
Parameter 1	R/W	-	-	-	SS_P ANEL	GS_P ANEL	REV_ PANEL	CFHR	-	-
Description	This command is used to panel related setting.									
	SS_Panel: Set Source scan output direction.									
	SS_Panel	S/D scan direction								
	0	S1 -> S240								
	1	S240-> S1								
	GS_Panel: Set Gate scan output direction.									
	GS_Panel	G/D scan direction								
	0	Top -> Bottom Scan (G1->G320)								
	1	Bottom -> Top Scan (G320 -> G1)								
	Note: Exmaple, resolution is 1080RGB x 1920.									
REV_Panel: Set the display of the same data on both normally-white and normally-black panels.										
REV_Panel	Panel (Normally-type)									
0	Normal Black panel									
1	Normal White panel									
CFHR: Set color filter horizontal alignment order.										
CFHR	Horizontal color filter alignment									
0	<S1><S2> alignment to <R1><G1><B1><R2><G2><B2> alignment									
1	<S1><S2> alignment to <B1><G1><R1><B2><G2><R2> alignment									
Restriction	-									

2.3.12.SET_BIST: BIST Pattern setting (Page0 - C2h)

[illegible]

2.3.13.SETRGBCYC: Set Display Waveform Cycles of RGB Mode (Page0 - C3h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX											
Command	R/W	1	1	0	0	0	0	1	1	C3											
Parameter 1	R/W		RGB_INV_PI[1:0]		RGB_INV_I[1:0]		IDLE_TYPE	RGB_INV_NP[1:0]													
Parameter 2	R/W					RGB_GND[3:0]															
Parameter 3	R/W				RGB_EQ1[4:0]																
Parameter 4	R/W				RGB_EQ2[4:0]																
Parameter 5	R/W	RGB_CHGEN_OFF[1:0]		RGB_CHGEN_ON[5:0]																	
Parameter 6	R/W	RGB_OFF[7:0]																			
Parameter 7	R/W	GAOPOFF_1ST_OFF	GAOPOFF_POL	GAOPOFFEN	GAOPOFF[4:0]																
Parameter 8	R/W	GAP2_ON[3:0]				SAP2_ON[3:0]															
Description	This command is used to set internal SD output timing.																				
	IDLE_TYPE: 0: (MSB,5'b0) 1: 6'b(MSB)																				
	RGB_INV_NP[1:0]: Set source dot inversion type at normal or partial mode. RGB_INV_PI[1:0]: Set source dot inversion type at (idle+partial) mode. RGB_INV_I[1:0]: Set source dot inversion type at idle mode.																				
	<table><tr><td>RGB_INV_NP[1:0] RGB_INV_PI[1:0] RGB_INV_I[1:0]</td><td>Dot inversion type</td></tr><tr><td>00</td><td>1-dot</td></tr><tr><td>01</td><td>2-dot</td></tr><tr><td>10</td><td>Column</td></tr><tr><td>11</td><td>inhibit</td></tr></table>										RGB_INV_NP[1:0] RGB_INV_PI[1:0] RGB_INV_I[1:0]	Dot inversion type	00	1-dot	01	2-dot	10	Column	11	inhibit	
	RGB_INV_NP[1:0] RGB_INV_PI[1:0] RGB_INV_I[1:0]	Dot inversion type																			
	00	1-dot																			
	01	2-dot																			
	10	Column																			
	11	inhibit																			
	RGB_GND[3:0]: Set positive source output GND function. RGB_EQ1[4:0]: Set negative SD EQ1 time. RGB_EQ2[4:0]: Set negative SD EQ2 time.																				
<table><tr><td>RGB_GND[3:0] RGB_EQ1[4:0] RGB_EQ2[4:0]</td><td>Clock cycle</td></tr><tr><td>00h</td><td>0 timing clock</td></tr><tr><td>01h</td><td>1 timing clock</td></tr><tr><td>:</td><td>1 timing clock /step</td></tr><tr><td>1Eh</td><td>30 timing clock</td></tr><tr><td>1Fh</td><td>31 timing clock</td></tr></table>										RGB_GND[3:0] RGB_EQ1[4:0] RGB_EQ2[4:0]	Clock cycle	00h	0 timing clock	01h	1 timing clock	:	1 timing clock /step	1Eh	30 timing clock	1Fh	31 timing clock
RGB_GND[3:0] RGB_EQ1[4:0] RGB_EQ2[4:0]	Clock cycle																				
00h	0 timing clock																				
01h	1 timing clock																				
:	1 timing clock /step																				
1Eh	30 timing clock																				
1Fh	31 timing clock																				
Note: 1 timing clock = 8 multiple of Fosc.																					
RGB_CHGEN_ON[5:0]: Set SD OP output time. RGB_CHGEN_OFF[7:0]: Set SD OP output time (after RGB_OFF).																					
<table><tr><td>RGB_CHGEN_OFF[7:0]</td><td>Clock cycle</td></tr><tr><td>00h</td><td>0 timing clock</td></tr><tr><td>01h</td><td>1 timing clock</td></tr><tr><td>:</td><td>1 timing clock /step</td></tr><tr><td>FEh</td><td>254 timing clock</td></tr><tr><td>FFh</td><td>255 timing clock</td></tr></table>										RGB_CHGEN_OFF[7:0]	Clock cycle	00h	0 timing clock	01h	1 timing clock	:	1 timing clock /step	FEh	254 timing clock	FFh	255 timing clock
RGB_CHGEN_OFF[7:0]	Clock cycle																				
00h	0 timing clock																				
01h	1 timing clock																				
:	1 timing clock /step																				
FEh	254 timing clock																				
FFh	255 timing clock																				
Note: 1 timing clock = 4 multiple of Fosc.																					

RGB_OFF[7:0]: Set SD data output time.

RGB_OFF[7:0]	Clock cycle
00h	0 timing clock
01h	1 timing clock
:	1 timing clock /step
FEh	254 timing clock
FFh	255 timing clock

Note: 1 timing clock = 4 multiple of Fosc.

GAOPOFF_1ST_OFF:

1:Enable
0:Disable

GAOPOFF_POL:

1:Enable
0:Disable

GAOPOFFEN:

1:Enable
0:Disable(GAOPOFF keeps low)

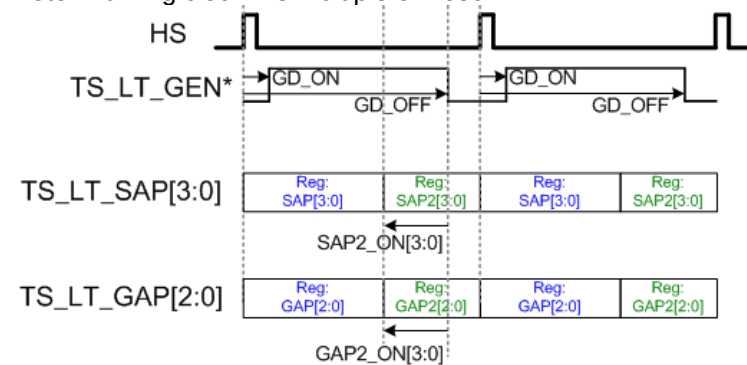
GAOPOFF [4:0]: Set GAMMA OP OFF time.

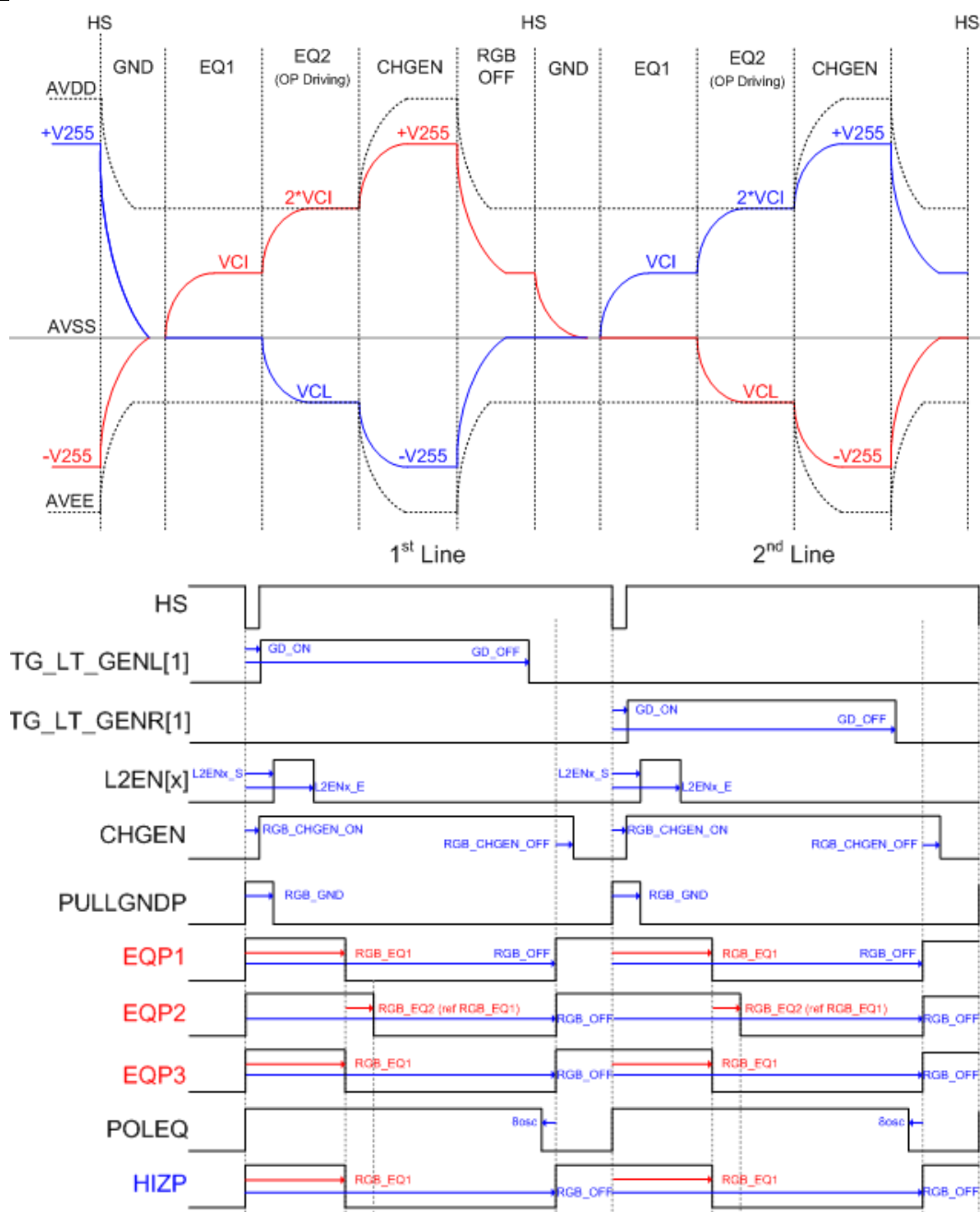
GAP2_ON[3:0]: setting Gamma OP output driving ability near gate off.

SAP2_ON[3:0]: setting channel OP output driving ability near gate off.

GAOPOFF[4:0] GAP2_ON[4:0] SAP2_ON[4:0]	Clock cycle
00h	0 timing clock
01h	1 timing clock
:	1 timing clock /step
1Eh	30 timing clock
1Fh	31 timing clock

Note: 1 timing clock = 8 multiple of Fosc.





Note:
1. If sets RGB_EQ*=00, EQ pulse will keep low.

Restriction

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2.3.14.SET TCON: Timing control setting (Page0 - C4h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	0	0	1	0	0	C4
Parameter 1	R/W	-	-	-	-	VBFP_RATIO[1:0]		TE_OPT[1:0]		-
Parameter 2	R/W	-	TE_DELAY[6:0]							-
Parameter 3	R/W	LN[7:0]							-	
Parameter 4	R/W	SLT_NP[7:0]							-	
Parameter 5	R/W	-	VFP_NP[6:0]							-
Parameter 6	R/W	-	VBP_NP[6:0]							-
Parameter 7	R/W	-	HBP_NP[6:0]							-
Parameter 8	R/W	SLT_I[7:0]							-	
Parameter 9	R/W	-	VFP_I[6:0]							-
Parameter 10	R/W	-	VBP_I[6:0]							-
Parameter 11	R/W	-	HBP_I[6:0]							-
Parameter 12	R/W	HBP_NCK[3:0]				HFP_NCK[3:0]				-
Description	This command is used to set timing control.									
	TE_OPT[0]: 1'b0: Enable TE during V-blanking. 1'b1: Disable TE during V-blanking at odd frames.									
	TE_OPT[1]: 1'b0: Normal TE. 1'b1: Fram_cnt[0], i.e., output 1'b1 at odd frames.									
	TE_Delay[6:0]: TE delay setting									
	TE_Delay[6:0]								Gate Line	
	0	0	0	0	0	0	0	0	4	
	0	0	0	0	0	0	0	1	8	
	-	-	-	-	-	-	-	-	4 line / step	
	0	1	0	1	0	0	0	0	320	
	LN[7:0]: Sets the gate line number to drive LCD panel. The number of lines must be equal or more than the gate line number of LCD panel. Gate line number = LN*2.									
LN[7:0]								Gate Line		
0	0	0	0	0	0	0	0	0		
0	0	0	0	0	0	0	1	2		
-	-	-	-	-	-	-	-	2 line / step		
1	0	1	0	0	0	0	0	320		
SLT_xx[7:0]: Sets the scan line time width. (4 x OSC) CLK / step										
STL_xx[7:0]								Scan Line time(OSC CLK)		
0	0	0	0	0	0	0	0	Inhibit		
0	0	0	0	0	0	0	1	(4x OSC) CLK		
0	0	0	0	0	0	1	0	(8 x OSC) CLK		
0	0	0		0	0	1	1	(12 x OSC) CLK		
-	-	-	-	-	-	-	-	(4 x OSC) CLK / step		
1	1	1	1	1	1	0	1	(1012 x OSC) CLK		
1	1	1	1	1	1	1	0	(1016 x OSC) CLK		
1	1	1	1	1	1	1	1	(1020 x OSC) CLK		
Note: fosc = 10MHz.										

VBP_xx[7:0]: Vertical back porch number setting.
VFP_xx[7:0]: Vertical front porch number setting.
HBP_xx[7:0]: Horizontal back porch number setting.(for DPI sync mode)

VBP_xx[6:0] VFP_xx[6:0] HBP_xx[6:0]	Number of VBP/ VFP(line) HBP(dot clock)
8h'00	Inhibit
8h'01	1
8h'02	2
:	:
8h'7E	126
8h'7F	127

Note: VBP = External (VS + VBP) -1 line. VFP = External VFP line.

XXX_NP[6:0]: for normal+partial mode

XXX_I[6:0]: for idle mode

VBFP_RATIO [1:0]: VFP and VBP ratio

RGB_INV_NP[1:0]	ratio
00	x1
01	x2
10	x4
11	inhibit

HFP_NCK[3:0]: Defined the number of HFP in the internal time.

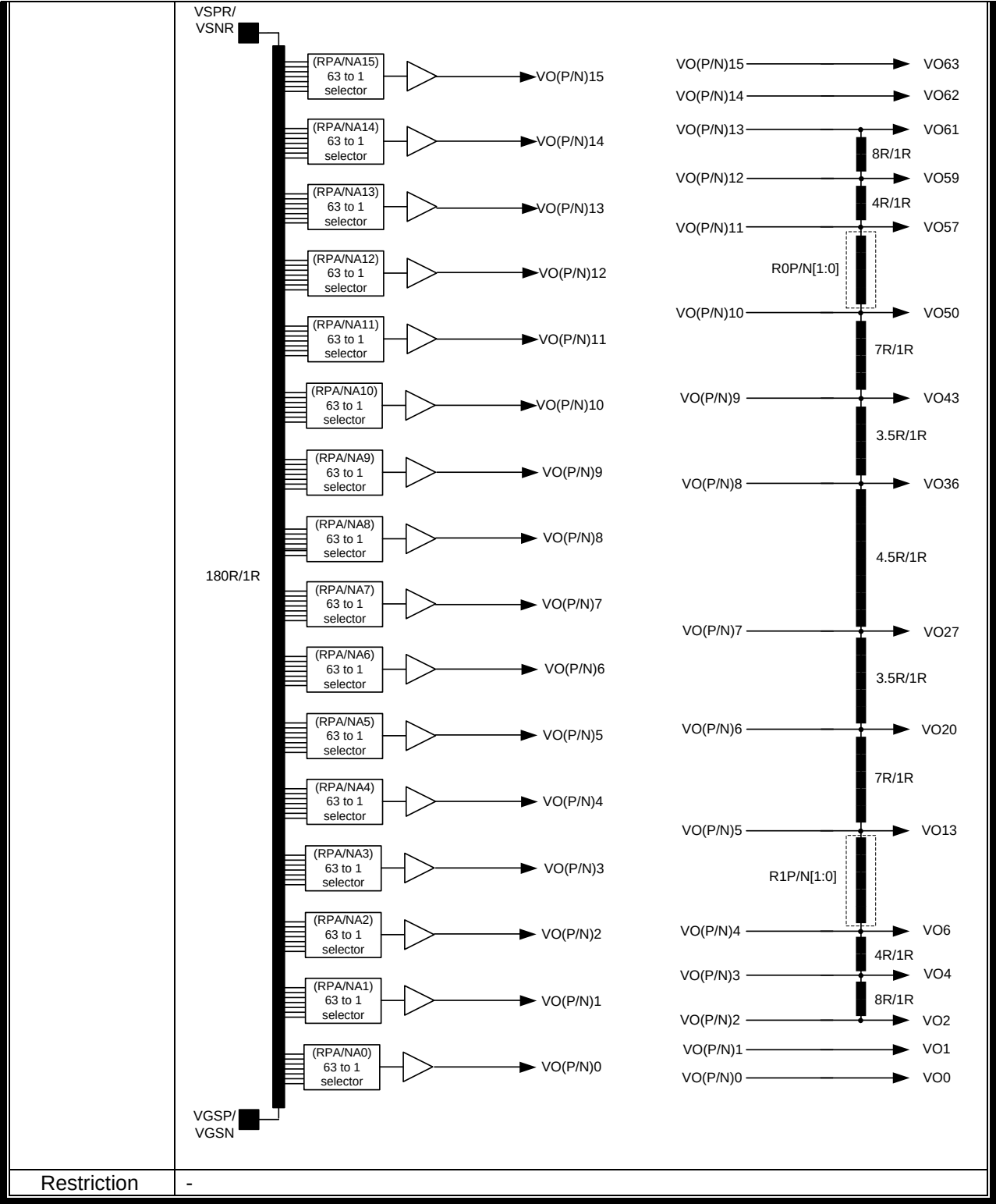
HBP_NCK[3:0]: Defined the number of HBP in the internal time.

Restriction

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2.3.15.SET_R GAMMA: Set Red Gamma output voltage(Page0 - C8h)

CMD/Pas	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	0	1	0	0	0	C8
Parameter 1	R/W	-	RPA15[5:0]							-
Parameter 2	R/W	-	RPA14[5:0]							-
Parameter 3	R/W	-	RPA13[5:0]							-
Parameter 4	R/W	-	RPA12[5:0]							-
Parameter 5	R/W	-	RPA11[5:0]							-
Parameter 6	R/W	-	RPA10[5:0]							-
Parameter 7	R/W	-	RPA9[5:0]							-
Parameter 8	R/W	-	RPA8[5:0]							-
Parameter 9	R/W	-	RPA7[5:0]							-
Parameter 10	R/W	-	RPA6[5:0]							-
Parameter 11	R/W	-	RPA5[5:0]							-
Parameter 12	R/W	-	RPA4[5:0]							-
Parameter 13	R/W	-	RPA3[5:0]							-
Parameter 14	R/W	-	RPA2[5:0]							-
Parameter 15	R/W	-	RPA1[5:0]							-
Parameter 16	R/W	-	RPA0[5:0]							-
Parameter 17	R/W	-	RNA15[5:0]							-
Parameter 18	R/W	-	RNA14[5:0]							-
Parameter 19	R/W	-	RNA13[5:0]							-
Parameter 20	R/W	-	RNA12[5:0]							-
Parameter 21	R/W	-	RNA11[5:0]							-
Parameter 22	R/W	-	RNA10[5:0]							-
Parameter 23	R/W	-	RNA9[5:0]							-
Parameter 24	R/W	-	RNA8[5:0]							-
Parameter 25	R/W	-	RNA7[5:0]							-
Parameter 26	R/W	-	RNA6[5:0]							-
Parameter 27	R/W	-	RNA5[5:0]							-
Parameter 28	R/W	-	RNA4[5:0]							-
Parameter 29	R/W	-	RNA3[5:0]							-
Parameter 30	R/W	-	RNA2[5:0]							-
Parameter 31	R/W	-	RNA1[5:0]							-
Parameter 32	R/W	-	RNA0[5:0]							-
Description	This command is used to set postive /neagative volatge of source output.									



2.3.16.IF_OPTION: Interface Option (Page0-CEh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	0	1	1	1	0	CE
Parameter 1	R/W	reserve								6A
Parameter 2	R/W	-	QSPI_SDO_TYP E[1:0]-		-	-	-	-	-	00
Description	QSPI_SDO_TYPE[1:0]:									
	QSPI_SDO_TYPE = 2'b11/2'b00 (default)			QSPI_SDO_TYPE = 2'b10			QSPI_SDO_TYPE = 2' b01			
	QSPI									
		CASE1		CASE2		CASE3				
		W	R	W	R	W	R			
	SDA(in/out)	input	output	input	input	input	input			
	SDO(out)	HiZ	HiZ	HiZ	output	HiZ	HiZ			
	DB[0](in/out)	input	input	input	input	input	output			
	Restriction	-								

2.3.17.SET_GIP_L: SET CGOUTx_L Signal Mapping, GS_Panel=0 (Page0-D0h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	1	0	0	0	0	D0
Parameter 1	R/W	-	-	-	-	-	GD_GAS_GATE_CLK	-	GD_LRSW	-
Parameter 2	R/W	-	-	GD_ON[5:0]						-
Parameter 3	R/W	GD_OFF[7:0]								-
Parameter 4	R/W	-	-		GD_EQ_PTR0[4:0]					-
Parameter 5	R/W	-	-	-	GD_EQ_PTR1[4:0]					-
Description	This command sets GIP CGOUTx_L signal mapping when XOR(GS_Panel, GS)=0. GD_GAS_GATE_CLK // 0: not gating clock in gas (default) // 1: gating clock in gas GD_LRSW gate driver L/R switch // 0: gate driver L (default) // 1: gate driver R GD_ON[5:0] gate driver on (Time unit = osc*4) GD_OFF[7:0] gate driver off (Time unit = osc*4) GD_EQ_PTR0[5:0] gate driver EQ0 pulse width(Time unit = osc*4) EQ0 end point = gate driver on + EQ0 pulse wodth GD_EQ_PTR1[7:0] gate driver EQ1 pulse width(Time unit = osc*4) EQ1 start point = gate driver off - EQ1 pulse wodth									
Restriction	-									

2.3.18.ABE_CTRL: (Page0-D3h)

[illegible]

2.3.19.RAMCTRL (Page0 - D7h)

[illegible]

2.3.20.AUTO_DISP_SETTING (Page0 - D8h)

[illegible]

2.3.21.OTP_PROG: OTP Program (Page0 - D9h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	1	1	0	0	1	D9
Parameter 1	R/W	OTP_INDEX[6:0]								-
Parameter 2	R/W	-	-	-	-	-	-	OTP_INT_VPP	OTP_AUTO_PROG	-
Parameter 3	R	OTP_RDATA[7:0]								-
Parameter 4	R/W	OTP_WDATA[7:0]								-
Parameter 5	R/W	-	-	-	-	-	OTP_RLD_PT M[1:0]		-	-
Parameter 6	R/W	OTP_PPROG	OTP_VPP_SEL	OTP_PRD	OTP_PWE	-	-	OTP_PTM[1:0]		-
Description	OTP_INDEX[6:0] : Set index of OTP table for programming. OTP_INT_VPP: Select VPP voltage from external or internal. 0: VPP voltage from external power supply. (VPP pad) (default) 1: VPP voltage from internal power supply (VGH). OTP_LOAD_DISABLE: Auto reload OTP value. 0: Enable auto reload OTP value after SLPOUT command send. (default) 1: Disable auto reload OTP value after SLPOUT command send. OTP_AUTO_PROG: When set to 1, the register content of OTP index is programmed. OTP_RLD_PTM[1:0]: for margin read.									
Restriction	-									

2.3.22.SET_WD: Setup watch dog (Page0 - DDh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	1	1	1	0	1	DD
Parameter 1	R/W	-	WD_OFF_F	WD_MODE[1:0]		FBLK_EN	WD_CLK_SEL[2:0]			-
Parameter 2	R/W	WD_Timer[7:0]								-
Parameter 3	R/W						ESD_TCON_OPT	ESD_SD_OPT	ESD_PRT_EN	-
Parameter 4	R	-					-	WD_FMS	WD_FBLK	-
Description	WD_MODE[1:0]: Watch dog mode select.									
	WD_MODE[1:0]		WD Alarm start		WD Alarm stop					
	0	0	Inhibit		Inhibit					
	0	1	Inhibit		Inhibit					
	1	0	Display off		Display on					
	1	1	Inhibit		Inhibit					
	WD_OFF: Control watch dog function on/off while external video signal was disappear. 0: Enable watch dog function. (default) 1: Disable watch dog function.									
	WD_CLK_SEL[2:0]: Set base clock frequency of watch dog.									
	WD_CLK_SEL[2:0]		WDosc (us)							
	000		0.5us							
	001		1us							
	010		2us							
	011		4us							
	100		8us							
	WD_Timer[7:0]: Set watch dog timer. When external HSYNC stop, watch dog timer start to count. Send alarm signal to watch dog if timer counting value is the same as WD_Timer[7:0] setting. When HSYNC resume, alarm signal is stopped.									
WD_Timer[7:0]		Time to alarm (ms)								
0x00		0 x WDosc(Inhibit)								
0x01		1 x WDosc								
...		1 WDosc / step								
0xFE		254 x WDosc								
0xFF		255 x WDosc								
Total duration to activate WD_Alarm = WD_TIMER[7:0]*Time Unit*4										
WD_FMS: Force Mode Switch when WDT occurs(not release)										
WD_FBLK: Force Blanking when WDT or Hwreset occurs(not release)										
ESD_SD_OPT: Define the source reaction while ESD 0:Floating 1:Keep the original data										
ESD_TCON_OPT: 1:If ESD happened in A line the display data will stop updating a frame. It will update new display data on next frame A+1 line. 0:IF ESD happened in A line the display data of A+1 line will same as A line.										
Restriction	-									

2.3.23.SET_PAGE(Page0 - DEh)

[illegible]

2.3.24.SET_PASSWORD: (Page0 - DFh)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	1	1	1	1	1	DF
Parameter 1	R/W	PASSWD1[7:0]								-
Parameter 2	R/W	PASSWD2[7:0]								-
Description	This command is used to set password to access inhouse register.									
	Normal mode:									
	{PASSWD1, PASSWD2, PASSWD3}				Description					
	{98h, 53h }				Standard & All inhouse CMD access					
	{5Ah, A5h }				Enable STD CMD and P0, P1 Registers					
	{09h, B1h}				Enter ESD protection mode					
	Others				Standard CMD access (default)					
Restriction	-									

2.3.25.UP_START_CTRL: Micro process control (Page0 - E8h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	1	0	1	0	0	0	E8
Parameter 1	R/W	UP_KEY1[7:0]								-
Description	UP_KEY1[7:0]: This command is used to enter OTP program mode. Before OTP programming, setting UP_KEY1=5Ah and UP_KEY2=A5h first. Otherwise OTP programming will fail.									
	UP_KEY1[7:0]		Description							
	0x5Ah		OTP control by up							
Restriction	-									

2.3.26.SETID: Set ID (Page0 - E9h)

[illegible]

2.4.1. SET_GPO: General Purpose Output setting(Page1 - BBh)

2.4.1. SET_GPO: General Purpose Output setting(Page1 - BBh)

[illegible]

	Ah	TO_LT_OTP_PPROG	TO_LT_OTP_PWE	TP_LT_OTP_VPP_SEL
	Bh	IFC__if_index_2e	IFC__if2ram_r_clk	IFC__if2ram_rde
	Ch	MIPID__2c3c_vs	MIPID__2c3c_pclk	MIPID__2c3c_de
	Dh	TS_LTL_GASHORT	TS_LT_GAOPOFF	TP_LT_EQ2
	Eh	TP_LT_EQ3	TCON__PWR_FM_CTRL	TCON__PWR_LN_CTRL
	Fh	DISP_sync[0]	DISP_sync[1]	REGS__dte_sample
Restriction	-			

2.4.2. GAMMA_POWER_TEST: Set GAMMA power related setting (Page1 - BEh)

[illegible]

2.4.3. SETRGBCYC2: Set RGB IF source switch control timing (Page1 - C1h)

[illegible]

2.4.4. SET GAMMA OP: Set Gamma OP related setting (Page1 - C4h)

[illegible]

2.4.5. SET_OSCM: Oscillator M setting (Page1 - C5h)

CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	0	0	1	0	1	C5
Parameter 1	R/W	-	-	OSC_DIV[1:0]		-	-	OSCM_ADJ[1:0]		-
Parameter 2	R/W	-	-	-	-	-	-	OSCM_EN	OSCM_BIAS_EN	-
Parameter 3	R/W	OSC_KEY[7:0]								
Description	This command is used to set oscillator M. OSCM (RC type) is used for HWRST & faster startup for MIPI LP mode.									
	OSCM_EN: Control oscillation of OSCM On/Off. 0 : Disable 1 : Enable									
	OSCM_BIAS_EN: Oscillator M bias current enable method. 0 : Bias turn on only when OSCM_EN=1 1 : Enables R path bias for fast start-up									
	OSCM_ADJ[1:0]: Adjust frequency oscillator M.									
	OSCM_ADJ[1:0]		OSCD (MHz) power		OSCD/2 (MHz)/2 TCON					
	0	0	16		8					
	0	1	20(default)		10					
	1	0	24		12					
	1	1	28		14					
	OSCM_DIV[1:0]: Adjust frequency oscillator M.									
	OSCM_ADJ[1:0]		TCON frequency							
	0	0	OSCD/2							
0	1	OSCD/4								
1	0	OSCD/8								
1	1	reserved								
OSC_KEY[7:0]: Set 5Ah to force disable Clock.										
Restriction	-									

2.4.6. SET INH CHKSUM: Set inhosue checksum(Page1 - D7h)

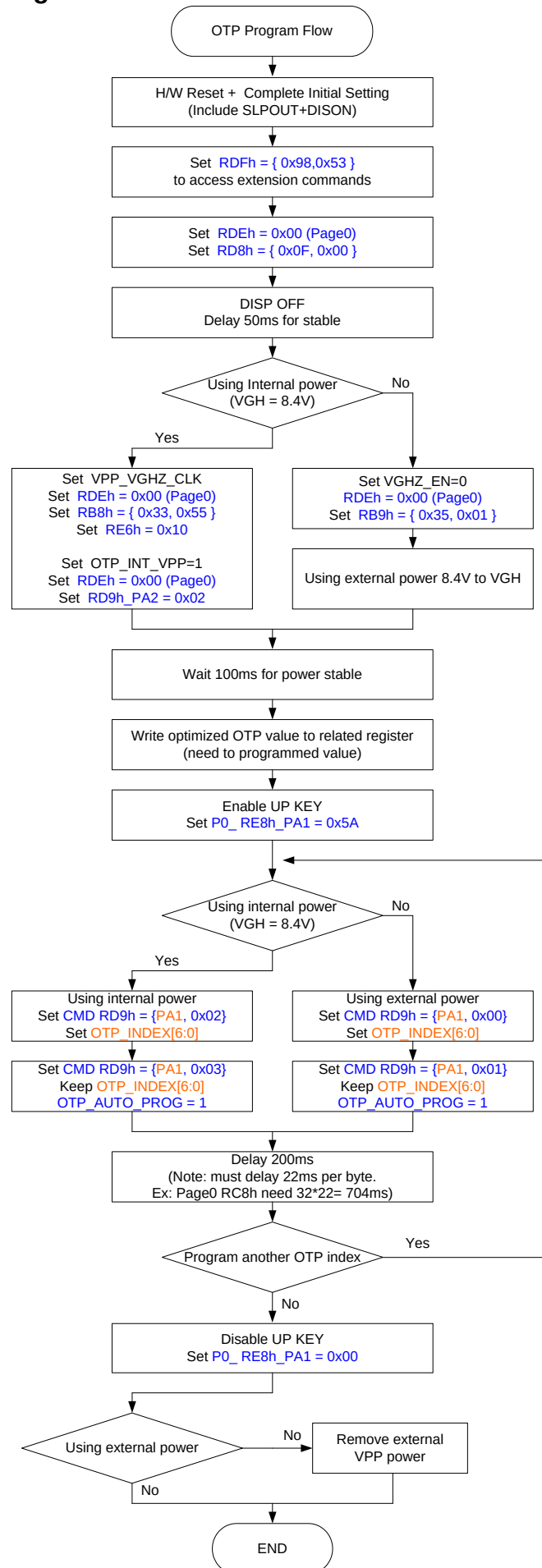
CMD/PAs	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	1	0	1	0	1	1	1	D7
Parameter 1	R/W	-	INH_CHKSUM_P AGE_SEL[1:0]	SLPIN_ RESET _EN	INH_C HKSU M_OFF	INH_CHKSUM_SEL[2:0]			-	
Description	This command is used to set checksum of inhouse register.									
	NH_CHKSUM_PAGE_SEL[1:0]: Select the inhouse checksum to be included in the calculation.									
	INH_CHKSUM_PAGE _SEL[1:0]		Pages included in calculation							
	00		Page0							
	01		Page0+Page1							
	10		Page0+Page1+Page2							
	11		Page0+Page1+Page2+Page3							
	SLPIN_RESET_EN:									
	0: Disable the function of reset Inhouse checksum in Sleep-in(for ESD TEST)									
	1: Enable the function of reset Inhouse checksum in Sleep-in									
Description	INH_CHKSUM_OFF:									
	0: Inhouse register checksum function ON									
	1: Inhouse register checksum function OFF									
	INH_CHKSUM_SEL[2:0]: The interval between each checksum calculation									
	INH_CHKSUM_SEL[2:0]:		Checksum Calculation Interval							
	000		2 Frames							
	001		4 Frames							
	010		8 Frames							
	011		16 Frames							
	100		32 Frames							
Others		64 Frames								
Restriction	-									

3.1. OTP table

OTP INDEX (HEX)	Ref. Command	Ref. Index	Duplicate	B7	B6	B5	B4	B3	B2	B1	B0
1	RAMCTRL	P0_D7_01	1	NVALID	CR_OPTIO	SPI_2LAN_	RP	RM	MLBIT_INV	DM[1:0]	
2		P0_D7_02		SCL_PL	EXT_VS_PL	EPF[1:0]		ENDIAN[1:0]		MDT[1:0]	
3		P0_C8_01		NVALID	-	RPA15[5:0]					
4	SET_R_GAMMA	P0_C8_02	1	-	-	RPA14[5:0]					
5		P0_C8_03		-	-	RPA13[5:0]					
6		P0_C8_04		-	-	RPA12[5:0]					
7		P0_C8_05		-	-	RPA11[5:0]					
8		P0_C8_06		-	-	RPA10[5:0]					
9		P0_C8_07		-	-	RPA9[5:0]					
A		P0_C8_08		-	-	RPA8[5:0]					
B		P0_C8_09		-	-	RPA7[5:0]					
C		P0_C8_10		-	-	RPA6[5:0]					
D		P0_C8_11		-	-	RPA5[5:0]					
E		P0_C8_12		-	-	RPA4[5:0]					
F		P0_C8_13		-	-	RPA3[5:0]					
10		P0_C8_14		-	-	RPA2[5:0]					
11		P0_C8_15		-	-	RPA1[5:0]					
12		P0_C8_16		-	-	RPA0[5:0]					
13		P0_C8_17		-	-	RNA15[5:0]					
14		P0_C8_18		-	-	RNA14[5:0]					
15		P0_C8_19		-	-	RNA13[5:0]					
16		P0_C8_20		-	-	RNA12[5:0]					
17		P0_C8_21		-	-	RNA11[5:0]					
18		P0_C8_22		-	-	RNA10[5:0]					
19		P0_C8_23		-	-	RNA9[5:0]					
1A		P0_C8_24		-	-	RNA8[5:0]					
1B		P0_C8_25		-	-	RNA7[5:0]					
1C		P0_C8_26		-	-	RNA6[5:0]					
1D		P0_C8_27		-	-	RNA5[5:0]					
1E		P0_C8_28		-	-	RNA4[5:0]					
1F		P0_C8_29		-	-	RNA3[5:0]					
20		P0_C8_30		-	-	RNA2[5:0]					
21		P0_C8_31		-	-	RNA1[5:0]					
22		P0_C8_32		-	-	RNA0[5:0]					
23		SETID		P0_E9_01	1	NVALID	-	-	-	-	-
24	P0_E9_02		ID1[7:0]								
25	P0_E9_03		ID2[7:0]								
26	P0_E9_04		ID3[7:0]								
27	P0_E9_05		ID4[7:0]								
28	P0_E9_01		2	NVALID	-	-	-	-	-	-	-
29	P0_E9_02			ID1[7:0]							
2A	P0_E9_03			ID2[7:0]							
2B	P0_E9_04			ID3[7:0]							
2C	P0_E9_05			ID4[7:0]							
2D	P0_E9_01		3	NVALID	-	-	-	-	-	-	-
2E	P0_E9_02			ID1[7:0]							
2F	P0_E9_03			ID2[7:0]							
30	P0_E9_04			ID3[7:0]							
31	P0_E9_05			ID4[7:0]							

32	GAMMA_SET	P0_B7_01	1	INVALID	VGSP_S[6:0]				
33		P0_B7_02		-	VGMP_S[6:0]				
34		P0_B7_03		-	VGSN_S[6:0]				
35		P0_B7_04		-	VGMN_S[6:0]				
36		P0_B7_01	2	INVALID	VGSP_S[6:0]				
37		P0_B7_02		-	VGMP_S[6:0]				
38		P0_B7_03		-	VGSN_S[6:0]				
39		P0_B7_04		-	VGMN_S[6:0]				
3A		P0_B7_01	3	INVALID	VGSP_S[6:0]				
3B		P0_B7_02		-	VGMP_S[6:0]				
3C		P0_B7_03		-	VGSN_S[6:0]				
3D		P0_B7_04		-	VGMN_S[6:0]				
48	VGPOW_SET	P0_B2_01	1	INVALID	VGPOW_OFFSET[6:0]				
49		P0_B2_01	2	INVALID	VGPOW_OFFSET[6:0]				
4A	SET_GAMMA_OP	P1_C4_01	1	INVALID	GAOPENN	GAOPENC	GAOPENP	R1P[1:0]	R0P[1:0]
4B		P1_C4_02		-	-	-	GAOP_CLA	R1N[1:0]	R0N[1:0]

3.2. OTP Programming Flow



3.3. Programming Sequence

Step	Operation
1	Power on and HWRST the module. Then do SLPOOUT.
2	Set 0xDFh = 0x98,0x51 to access extension commands.
3	Set page0 CMD(RDEh=0x00h) , 1) Internal power mode, set Page0 RD9h_PA2=0x02 (OTP_INT_VPP = 1) 2) External power mode, set CMD RB9h={0x35, 0x01}, then feed external power 8.4 to VGH. (OTP_INT_VPP = 0).
4	Wait 100ms for VPP power stable.
5	Write optimized OTP value to related register.
6	Set Page0 CMD(RDEh=0x00h), then set Page0_0xE8h = 0x5A to enable UP KEY.
7	Set 0xD9h_PA1 (OTP_INDEX[6:0]) that you want to programmed.
8	Set 0xD9h_PA2[0]=1 (OTP_AUTO_PROG=1)
9	Delay 200ms for internal power to program OTP. Or delay 200ms for external power to program OTP.
10	One OTP block programming was completed. Return to step (7) for next OTP block programming. Or go next step if all OTP programming is finished.
11	Set Page0_0xE8h = 0x00 to disable OTP KEY.
12	If you use external power, remove the external power 8.4V from VPP.

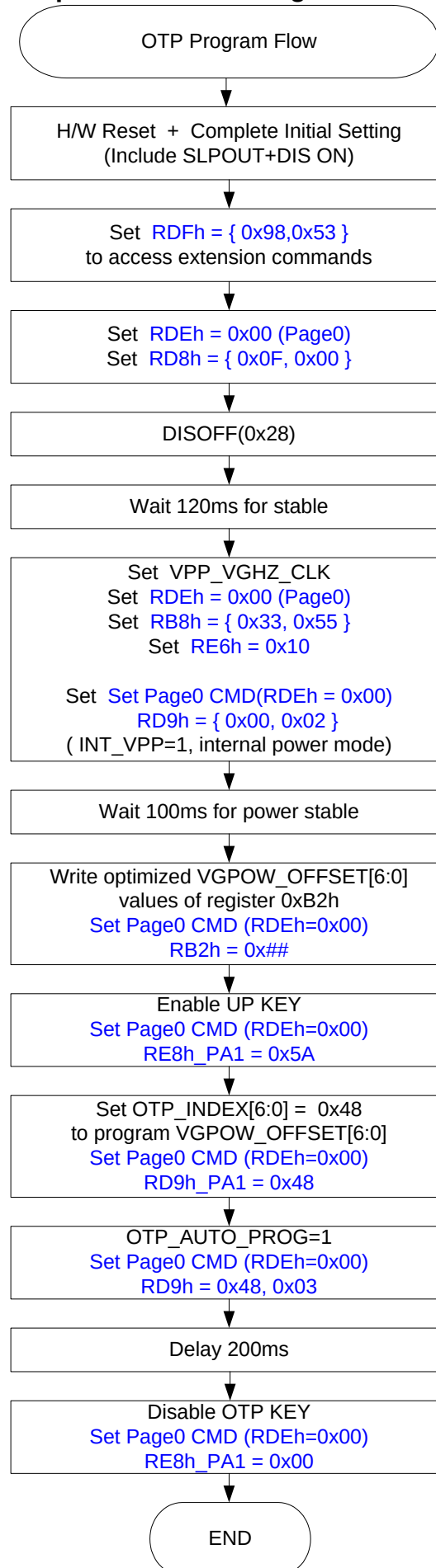
Note:

1. All OTP default value is 1.
2. OTP can be reloaded only NVALID bit is burned(value is 0).
3. Fifipower driver IC do auto OTP programming. User only set the start OTP_INDEX of the Ref. CMD block then all block is programmed by driver IC. This function also can be used for MTP Ref. CMD block.

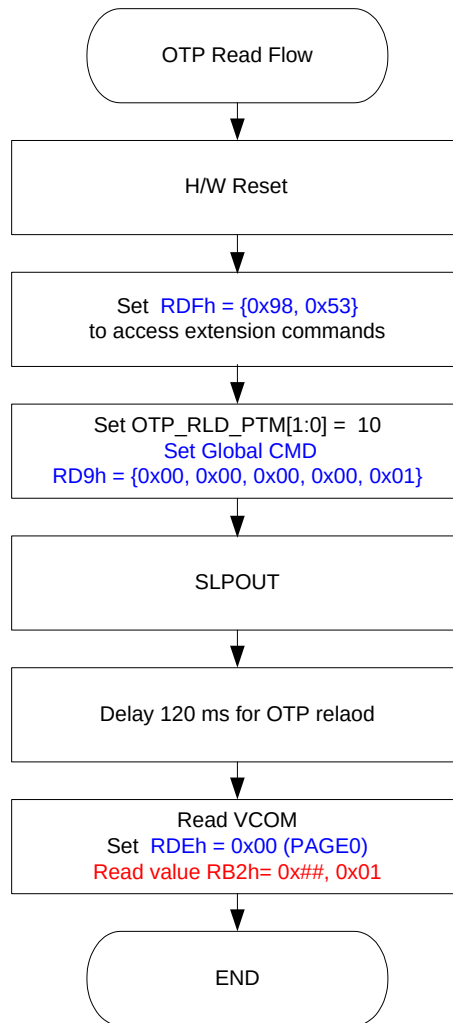
For example:

- a. To program VCOM_SET 1st time, user set OTP_INDEX=006h then from OTP_INDEX=006h to OTP_INDEX=007h(Ref. CMD block of 1st VCOM_SET) is programmed by driver IC.
- b. To program VCOM_SET 2nd time, user still set OTP_INDEX=006h then from OTP_INDEX=008h to OTP_INDEX=009h(Ref. CMD block of 2nd VCOM_SET) is programmed by driver IC.

3.4. OTP Programming example of VCOM setting

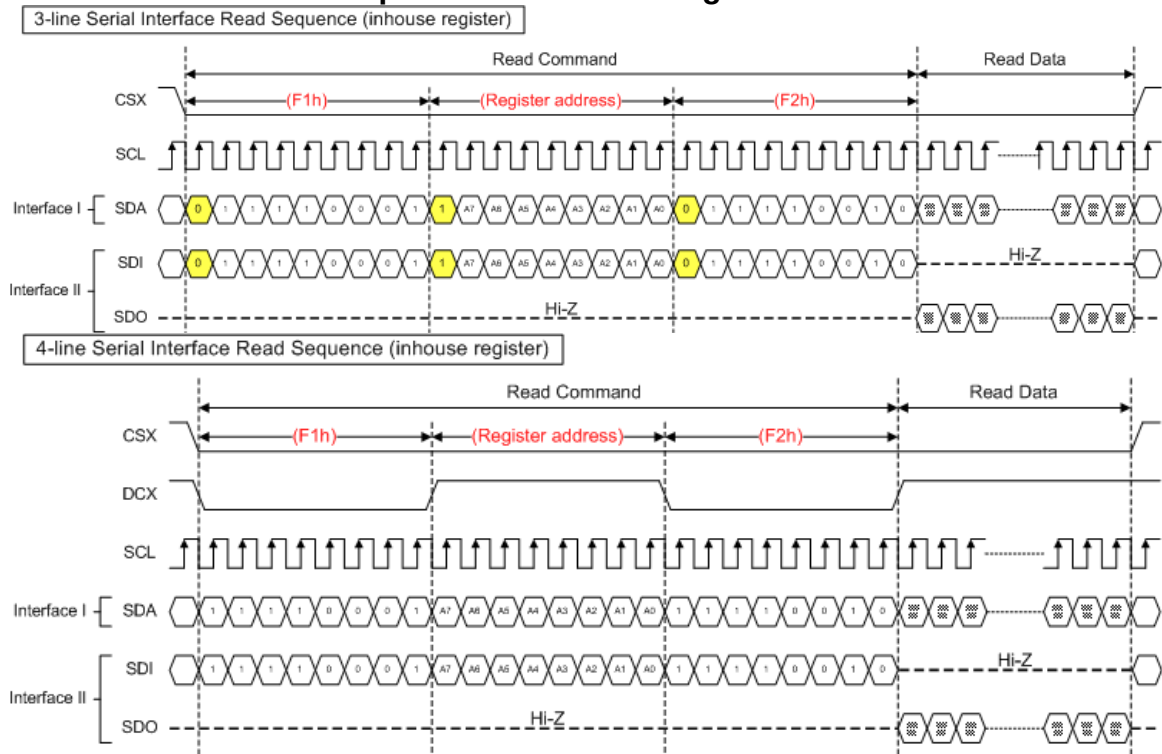


3.5. OTP read example



4. Read sequence for inhouse register

4.1. Serial interface read sequence for inhouse register



4.2. Parallel interface read sequence for inhouse register

